

Optimization

A Journal of Mathematical Programming and Operations Research

ISSN: 0233-1934 (Print) 1029-4945 (Online) Journal homepage: www.tandfonline.com/journals/gopt20

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To cite this article: Ouayl Chadli, Xin Li, Ram N. Mohapatra & Jen-Chih Yao (08 Oct 2025): A smoothing approximation approach to dynamical inertial newton systems for non-smooth and non-convex optimization: the deterministic case, Optimization, DOI: [10.1080/02331934.2025.2569782](https://doi.org/10.1080/02331934.2025.2569782)

To link to this article: <https://doi.org/10.1080/02331934.2025.2569782>



Published online: 08 Oct 2025.



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A smoothing approximation approach to dynamical inertial newton systems for non-smooth and non-convex optimization: the deterministic case

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ABSTRACT

Dynamical systems have inspired and explained several accelerated algorithms for a wide range of optimization problems. But due to the lack of smoothness and convexity of the objective functions in many real world applications, we cannot directly apply these accelerated algorithms in these situations. This paper proposes to apply a smoothing approximation approach to address non-smooth, non-convex machine learning optimization problems. Our work is motivated by the following goal: developing a direct method for finding critical points of objective functions of machine learning problems where functions are known to be non-smooth and non-convex. To achieve these goals, we establish the convergence of an alternative algorithm for smooth functions without convexity that supplements some recent results of Attouch et al.

ARTICLE HISTORY

Received 26 March 2025

Accepted 16 September 2025

KEYWORDS

Dynamical systems;
smoothing approximation;
optimization; ReLU function

2020 MATHEMATICS SUBJECT

CLASSIFICATIONS

37N40; 49M15; 90C26;
90C30; 65K05

1. Introduction

Polyak [1, Theorem 9] initiated the approach of using dynamical systems in speeding up the convergence of the gradient methods in optimization. This approach has been promoted and further extended by Attouch, Alvarez, and their collaborators since the late 90's (see, e.g. [2–6]). These important works allow us to better understand many accelerated algorithms and greatly facilitate further speedups of the convergence of existing iterative methods. A breakthrough came when Su, Boyd, and Candes (see [7]) applied this approach with the so-called averaged vanishing damping (AVD) system that can interpret Nesterov's accelerated gradient method [8] as a discretization in the time variable. For the latest development, please see [9, 10], and the references therein. The following continuous-time dynamical system was introduced in [5], referred to as Dynamical Inertial Newton (DIN) system:

$$\ddot{\theta}(t) + \alpha \dot{\theta}(t) + \beta \nabla^2 f(\theta(t)) \dot{\theta}(t) + \nabla f(\theta(t)) = 0, \quad \text{for } t \in (0, \infty) \quad \text{and} \quad \theta(0) = x_0, \quad (1)$$

where t is the time parameter, f is a function assumed to be of class $\mathcal{C}^2$, with gradient ∇f and Hessian $\nabla^2 f$, $\alpha > 0$, $\beta > 0$, and x_0 are parameters, and $\theta : \mathbb{R}_+ \rightarrow \mathbb{R}^p$. It was shown that, among other things, if $\theta(t)$ is a solution to (1), then, as $t \rightarrow \infty$, $f(\theta(t)) \rightarrow \min f$. That is, solutions to (1) converge to the minimizers of f . But there are several challenges in directly applying the results of the investigation of (1) in real world applications (such as machine learning) since the objective function f appears there may be nonsmooth or nonconvex. To overcome the explicit use of the Hessian $\nabla^2 f$, [5] introduced an auxiliary function $\psi : \mathbb{R}_+ \rightarrow \mathbb{R}^p$ and considered the following first order dynamical system (defined for f merely differentiable):

$$\begin{cases} \dot{\theta}(t) + \beta \nabla f(\theta(t)) + \left(\alpha - \frac{1}{\beta}\right) \theta(t) + \frac{1}{\beta} \psi(t) = 0 \\ \dot{\psi}(t) + \left(\alpha - \frac{1}{\beta}\right) \theta(t) + \frac{1}{\beta} \psi(t) = 0 \end{cases} \quad \text{for } t \in (0, \infty) \quad (2)$$

Indeed, letting $\psi(t) = -\beta \dot{\theta}(t) - \beta^2 \nabla f(\theta(t)) - (\alpha\beta - 1)\theta(t)$ we can re-write (1) as (2). Conversely, we can differentiate the first line of (2) and then eliminate $\dot{\psi}(t)$ using the second line to return to (1). Thus, when f is smooth, (1) is equivalent to (2). Note that (2) does not require the existence of second-order derivatives.

Castera et al. [11] was the first to adapt DIN to machine learning problems. To further overcome the computational and conceptual difficulties raised by $\dot{\theta}(t)$ and $\nabla f(\theta(t))$, the first step is to replace ∇f with the Clarke subdifferential ∂f , as carried out in [11] for the system (2), resulting in the study of the following first-order system of differential inclusion:

$$\begin{cases} \dot{\theta}(t) + \beta \partial f(\theta(t)) + \left(\alpha - \frac{1}{\beta}\right) \theta(t) + \frac{1}{\beta} \psi(t) \ni 0 \\ \dot{\psi}(t) + \left(\alpha - \frac{1}{\beta}\right) \theta(t) + \frac{1}{\beta} \psi(t) = 0 \end{cases} \quad \text{for } t \in (0, \infty) \quad (3)$$

Castera et al. [11] then presented a discretization of (3) to propose a new iterative algorithm. But their analysis of the algorithm only guaranteed the converge to the so-called D-critical points of f , a notion introduced in [11] which is, in general, a proper superset of the classical critical points. The cause of this compromise of using D-critical points, instead of the classical ones, is the fact that subdifferentials do not satisfy the sum rule (see more details below).

In this paper, we propose a different approach in order to avoid both the use of D-critical points and the computation of subdifferentials. To achieve our goals, we make two changes. First, we find it more convenient to use the following equivalent first order system for (1) in place of (2):

$$\begin{cases} \dot{\theta}(t) = -\alpha\theta(t) + \varphi(t) - \beta \nabla f(\theta(t)) \\ \dot{\varphi}(t) = -\nabla f(\theta(t)) \end{cases} \quad \text{for } t \in (0, \infty). \quad (4)$$

Secondly, when numerically solving (4) for the non-differentiable function $f(\theta)$, we use its smooth approximations, $S(\theta, \varepsilon)$. More precisely, we consider $f(\theta) \approx S(\theta, \varepsilon)$ as $\varepsilon \rightarrow 0^+$ (for more precise statement, please see Sections 3 and 4), and we first solve for a sequence

of $\varepsilon = \varepsilon_k$ with $\varepsilon_k \rightarrow 0^+$

$$\begin{cases} \dot{\theta}(t) = -\alpha\theta(t) + \varphi(t) - \beta\nabla S(\theta(t), \varepsilon) \\ \dot{\varphi}(t) = -\nabla S(\theta(t), \varepsilon) \end{cases} \quad \text{for } t \in (0, \infty); \quad (5)$$

and then we follow a ‘diagonal’ path to obtain a sequence of solutions at increasing time marks that yield approximations to the minimizers of f itself. The more precise statements will be given by Algorithm 2 with the proof of convergence given by Theorems in Section 4.

This paper is organized as follows. In Section 2, we provide the background and motivation for our study. In Section 3, we briefly review the important properties of the smoothing techniques developed in [12]. Finally, in Section 4, we present our algorithm, Smoothing Inertial Newton Algorithm (SINA), based on smoothing approximations and DIN systems and establish its convergence analysis.

2. Motivations and preliminaries

In this section, we provide some detailed accounts of the various difficulties and challenges that one faces in solving optimization problems that appear in machine learning. To better understand the specifics on the objective functions treated in machine learning applications, we will give a brief introduction on these background materials in Subsection 2.1 (DNN and its form of loss functions that are to be minimized) and Section 3 (the multilayer structures and the use of ReLU and other activation functions).

2.1. Deep neural network

Deep Neural Networks (DNNs) are the core of deep learning, a subfield of machine learning that empowers machines to learn complex patterns from data and perform tasks traditionally requiring human intelligence. DNNs have enabled machine learning’s success by efficiently learning accurate functional relationships between input and output data in training datasets. In a DNN, one searches for a functional relationship of a general type represented by a function $g : \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}^d$ such that $\theta \in \mathbb{R}^p \mapsto g(x, \theta)$ is locally Lipschitz continuous for each $x \in \mathbb{R}^m$. The parameters of the model are represented by the vector $\theta \in \mathbb{R}^p$, while $x \in \mathbb{R}^m$ denotes the input vector and $y := g(x, \theta) \in \mathbb{R}^d$ represents the output of the DNN. The ‘deep’ in DNN signifies the processing of input x through multiple layers of function composition, often visualized as a network diagram (see, e.g. [13, 14]). For our purpose, we could view g as a function with a set of parameters (contained in vector θ) that need to be ‘learned’ from a set of training data (for more details of the representation of g , please see Section 3). More precisely, we consider a data set of N samples $\{(x_n, y_n) : n = 1, 2, \dots, N\}$. To learn the parameter θ , we minimize a loss function over N training samples. Specifically, we seek θ that minimizes the discrepancy between the model’s output $g(x_n, \theta)$ and the true value y_n for each input x_n . Let $l(g(x_n, \theta), y_n)$ denote the loss incurred for a single sample. This is achieved by minimizing the following objective:

$$\min_{\theta \in \mathbb{R}^p} f(\theta) \quad (6)$$

where $f : \mathbb{R}^p \rightarrow \mathbb{R}$ denotes the total loss function given by

$$f(\theta) = \sum_{n=1}^N l(g(x_n, \theta), y_n), \quad \text{for any } \theta \in \mathbb{R}^p. \quad (7)$$

For convenience, we define $f_n(\theta) := l(g(x_n, \theta), y_n)$, representing the loss for the n th sample. Thus, the objective function can be written as $f(\theta) = \sum_{n=1}^N f_n(\theta)$. We face several challenges: (1) the model g may lack differentiability; (2) the per-sample loss function $f_n : \mathbb{R}^d \rightarrow \mathbb{R}$ is assumed to be at best locally Lipschitz continuous; and (3) the overall objective function f may be non-convex.

2.2. The non-differentiability and non-convexity

As discussed in the Introduction and Section 2.1, the objective functions we aim to minimize in deep learning are often neither differentiable nor convex. This poses significant optimization challenges. A powerful tool for handling non-differentiable functions is Clarke's subdifferential (see [15]). Instead of the traditional gradient, which may not exist, we employ the subdifferential ∂f . This leads to a shift from equalities to inclusions, as exemplified in (3).

However, solving differential inclusions of this type is far from trivial [11]. The transition from differential equations to differential inclusions introduces considerable complexity. For instance, analyzing the existence and uniqueness of solutions becomes considerably more delicate. Furthermore, developing efficient numerical methods for solving differential inclusions requires specialized techniques. Section 2.4 provides the necessary background definitions and properties of Clarke's subdifferential needed for further discussion.

Another significant obstacle when working with machine learning optimization problems with subdifferentials is that we have to deal with $f = \sum_{n=1}^N f_n$ (as given in (7)) and we need to calculate ∂f in terms of ∂f_n 's. Unfortunately, as explained in next subsection, in general, there is no 'sum rule'.

2.3. The lack of the sum rule of subdifferentials

The computation of subdifferentials and the convergence analysis of numerical optimization algorithms for problems like (6), where the objective $f = \sum_{n=1}^N f_n$, fundamentally depend on the sum rule for subdifferentials. Ideally, we would have $\partial f = \sum_{n=1}^N \partial f_n$. However, as is well known, the subdifferential of a sum is only guaranteed to be a subset of the sum of the subdifferentials. While equality holds under certain conditions (see, e.g. [15, Proposition 2.3.3 and Corollary 3] for a more general statement).

Proposition 2.1: *Assume that $f_1, f_2 : \mathbb{R}^p \rightarrow \mathbb{R}$ are locally Lipschitz. Then*

$$\partial (f_1 + f_2)(x) \subseteq \partial f_1(x) + \partial f_2(x). \quad (8)$$

Furthermore, if one of f_1 and f_2 is regular at x , then the equality holds in (8).

Unfortunately, the sum rule (i.e. the equality in (8)) does not hold in general in the non-convex setting using the standard Clarke's subdifferentials. For example take

$\varphi(x) = \varphi_1(x) + \varphi_2(x)$, where $\varphi_1(x) = |x|$ and $\varphi_2(x) = -\varphi_1(x)$, we have $\partial\varphi(0) = \{0\}$, while $\partial\varphi_1(0) + \partial\varphi_2(0) = [-1, 1] + [-1, 1] = [-2, 2]$. In practice, however, the limitations of the subdifferential sum rule are often overlooked in deep learning. This can lead to the identification of spurious stationary points that are not true Clarke critical points [11]. To address this issue, [11] introduced the more general notion of a D-critical point to be introduced next (see the paragraph after (9)).

2.4. Preliminaries

Recall that a function $\varphi : \mathbb{R}^p \rightarrow \mathbb{R}$ is locally Lipschitz continuous, if for any $\theta \in \mathbb{R}^p$, there exists a neighbourhood $\mathcal{N}$ of θ and a constant $c_{\mathcal{N}} > 0$ such that

$$|\varphi(\theta_1) - \varphi(\theta_2)| \leq c_{\mathcal{N}} \|\theta_1 - \theta_2\|, \quad \text{for any } \theta_1, \theta_2 \in \mathcal{N},$$

where $\|\cdot\|$ is the norm of $\mathbb{R}^p$. A function $\Phi : \mathbb{R}^p \rightarrow \mathbb{R}^d$ is locally Lipschitz continuous if each of its coordinates is locally Lipschitz continuous. By Rademacher's theorem (see, e.g. 16, p. 101–102), locally Lipschitz continuous functions $\varphi : \mathbb{R}^p \rightarrow \mathbb{R}$ are differentiable almost everywhere.

We recall the definition of generalized directional derivative and Clarke subdifferential for a locally Lipschitz function.

Definition 2.2: Let $\varphi : \mathbb{R}^p \rightarrow \mathbb{R}$. The generalized directional derivative of φ at $x \in \mathbb{R}^p$ in the direction $v \in \mathbb{R}^p$, denoted by $\varphi^0(x; v)$, is defined by

$$\varphi^0(x; v) = \limsup_{y \rightarrow x, t \rightarrow 0^+} \frac{\varphi(y + tv) - \varphi(y)}{t}.$$

When φ is a locally Lipschitz function, it is easy to see that $\varphi^0(x, v)$ is finite for all x and v . The Clarke subdifferential of φ at x , denoted by $\partial\varphi(x)$, is given by

$$\partial\varphi(x) = \{\zeta \in \mathbb{R}^p : \varphi^0(x; v) \geq \langle \zeta, v \rangle, \text{ for all } v \in \mathbb{R}^p\}.$$

Definition 2.3: A locally Lipschitz function $\varphi : \mathbb{R}^p \rightarrow \mathbb{R}$ is said to be regular at $x \in \mathbb{R}^p$ if for all $v \in \mathbb{R}^p$ the directional derivative $\varphi'(x; v)$ exists and

$$\varphi'(x; v) = \varphi^0(x; v).$$

Proposition 2.4 ([15, Theorem 2.5.1]): Let $\varphi : \mathbb{R}^p \rightarrow \mathbb{R}$ be a locally Lipschitz function. Then

$$\partial\varphi(x) = \text{conv}\{\zeta \in \mathbb{R}^p : \zeta = \lim_{k \rightarrow \infty} \nabla\varphi(x_k), x_k \rightarrow x, \varphi \text{ is differentiable at } x_k\},$$

where 'conv' denotes the convex hull.

Definition 2.5: A point $x \in \mathbb{R}^p$ is called a critical point of φ if it satisfies the condition $0 \in \partial\varphi(x)$.

Theorem 2.6 ([15, Proposition 2.3.2 and Proposition 2.1.5(a)]): Let $\varphi : \mathbb{R}^p \rightarrow \mathbb{R}$ be a locally Lipschitz function. If x is a local minimizer of the function φ , then

- (i) $\varphi^0(x; v) \geq 0$ for all $v \in \mathbb{R}^p$;
- (ii) $0 \in \partial\varphi(x)$.

3. Approximation by smoothing functions

Finding critical points $\bar{\theta}$ of the function $f(\theta) = \sum_{n=1}^N f_n(\theta)$ given by (7), i.e. $\bar{\theta} \in \mathbb{R}^p$ such that $0 \in \partial f(\bar{\theta})$, requires us to compute or approximate the subdifferential of f at each iteration as is done in [11]. Even the numerical algorithm can find $\bar{\theta}$ such that

$$0 \in \sum_{n=1}^N \partial f_n(\bar{\theta}), \quad (9)$$

unfortunately, the lack of sum rule (cf. Section 2.3) means that we could not assert that $0 \in \partial f(\bar{\theta})$ in general. This motivated the introduction of D-critical point in [11]: the collection of all points satisfying (9).

We will focus on finding the classical (Clarke's) critical points of the original, potentially non-smooth function f . Our approach centres on solving (9) by leveraging smoothing approximations of f . Crucially, these approximations replace subdifferentials with gradients, thereby guaranteeing the applicability of the sum rule. We then construct an algorithm that provably converges to a critical point of f itself. We concentrate specifically on the ReLU activation function, the most prevalent choice for g in deep learning.

In this section, we will first review the theory of smoothing approximations and then show how smoothing approximations can be constructed for functions used in machine learning.

Definition 3.1: Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ be a continuous function. We call $S : \mathbb{R}^p \times \mathbb{R}_+ \rightarrow \mathbb{R}$ a smoothing function of f , if

- (i) For any fixed $\varepsilon > 0$, $S(\theta, \varepsilon)$ is continuously differentiable in $\theta \in \mathbb{R}^p$;
- (ii) For any $\theta \in \mathbb{R}^p$, $\lim_{\omega \rightarrow \theta, \varepsilon \downarrow 0} S(\omega, \varepsilon) = f(\theta)$;
- (iii) For any $\theta \in \mathbb{R}^p$, $\{\zeta \in \mathbb{R}^p : \nabla_{\theta} S(\theta_k, \varepsilon_k) \rightarrow \zeta, \text{ for some } \theta_k \rightarrow \theta, \varepsilon_k \downarrow 0\} \subseteq \partial f(\theta)$

The subdifferential associated with the smoothing function S of f is given by

$$G_S(\theta) := \text{conv}\{\zeta \in \mathbb{R}^p : \nabla_{\theta} S(\theta_k, \varepsilon_k) \rightarrow \zeta, \text{ for some } \theta_k \rightarrow \theta, \varepsilon_k \downarrow 0\}. \quad (10)$$

The *gradient consistency* condition

$$G_S(\theta) \subset \partial f(\theta), \quad \text{for all } \theta \in \mathbb{R}^p \quad (11)$$

between the Clarke subdifferential of the locally Lipschitz function f and the subdifferential associated with the smoothing function S of f plays an important role for the convergence of smoothing approximations.

The following proposition gives important properties of the compositions of smoothing functions.

Proposition 3.2 ([17]): (i) If $S_1, \dots, S_N$ are smoothing functions of $f_1, \dots, f_N$, respectively, then $S := \sum_{i=1}^N S_i$ is a smoothing function of $f := \sum_{i=1}^N f_i$ when f_i is regular for every $i = 1, 2, \dots, N$.

- (ii) Let $l : \mathbb{R} \rightarrow \mathbb{R}$ be continuously differentiable and globally Lipschitz, and $g : \mathbb{R}^p \rightarrow \mathbb{R}$ be locally Lipschitz. If S is a smoothing function of g , then $l \circ S$ is a smoothing function of $l \circ g$.
- (iii) Let $l : \mathbb{R} \rightarrow \mathbb{R}$ be regular, and $g : \mathbb{R}^m \rightarrow \mathbb{R}^p$ be continuously differentiable. If S is a smoothing function of l , then $S \circ g$ is a smoothing function of $l \circ g$.

It has been shown in [18] that for any locally Lipschitz function f , we can construct a smoothing function by using convolution

$$S(\theta, \varepsilon) = \int_{\mathbb{R}^p} f(\theta - s) \rho_\varepsilon(s) \, ds = \int_{\mathbb{R}^p} f(s) \rho_\varepsilon(\theta - s) \, ds$$

where $\rho_\varepsilon : \mathbb{R}^p \rightarrow \mathbb{R}$ is a smooth kernel function, such that the gradient consistency condition (11) holds.

Using the smoothing function S and $\nabla_\theta S$, [17] provided a smoothing approximation approach to solve a minimization problem as follows:

Step 1 : Define a smoothing function $S : \mathbb{R}^p \times \mathbb{R}_+ \rightarrow \mathbb{R}$ to approximate f .

Step 2 : Use an algorithm to find an approximate solution of the smooth optimization problem:

$$\min_{\theta \in \mathbb{R}^p} S(\theta, \varepsilon_k) \quad (12)$$

for any $\varepsilon_k > 0$.

Step 3 : Update ε_k to guarantee the convergence of the smoothing approximation to a local minimizer or a stationary point of the nonsmooth optimization problem (6).

Since we are interested in smooth approximations in deep learning, we focus our attention on smooth approximations of non-smooth functions formulated by a composition of the ReLU, or the plus function $(t)_+ := \max\{0, t\}$. We follow [12]: let $\rho : \mathbb{R} \rightarrow \mathbb{R}_+$ be a piecewise continuous function satisfying

$$\int_{-\infty}^{\infty} \rho(s) \, ds = 1, \quad \rho(s) = \rho(-s), \quad \text{and} \quad r_\rho := \int_{-\infty}^{\infty} |s| \rho(s) \, ds < \infty \quad (13)$$

and define $P_\rho : \mathbb{R} \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ by

$$P_\rho(t, \varepsilon) := \int_{-\infty}^{\infty} (t - \varepsilon s)_+ \rho(s) \, ds < \infty. \quad (14)$$

From (13), the function $P_\rho(\cdot, \cdot)$ is well defined. Furthermore, for any fixed $\varepsilon > 0$, the function $P_\rho(\cdot, \varepsilon)$ is convex, strictly increasing, and continuously differentiable, with $\nabla_t P_\rho(t, \varepsilon) = \int_{-\frac{t}{\varepsilon}}^{\frac{t}{\varepsilon}} \rho(s) \, ds$. Moreover, P_ρ satisfies the following inequality

$$0 \leq P_\rho(t, \varepsilon) - (t)_+ \leq \varepsilon r_\rho. \quad (15)$$

It follows that for any $t \in \mathbb{R}$, we have

$$\lim_{z \rightarrow t, \varepsilon \downarrow 0} P_\rho(z, \varepsilon) = (t)_+.$$

Hence, from Definition 3.1 and choice of P_ρ , the following result would imply that P_ρ is indeed a smoothing function of the plus function $(t)_+$ associated with the density ρ .

Lemma 3.3 ([17]): *The function P_ρ given by (14) is a smoothing function of the plus function $(t)_+$ satisfying the gradient consistency condition*

$$G_{P_\rho}(t) = \partial(t)_+,$$

where $\partial(t)_+$ is the Clarke subdifferential of the plus function and $G_{P_\rho}(\cdot)$ is the subdifferential associated with the smoothing function P_ρ .

Now, with applications to deep learning in mind, we consider a non-smooth function $g : \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$ defined by a nested compositions of affine functions composed with the plus (ReLU) function:

$$g(x, \theta) = W_d(\cdots (W_2(W_1x + b_1)_+ + b_2)_+ \cdots)_+ + b_d, \quad (16)$$

where W_i 's are matrices and b_i 's are scalars for $i = 1, \dots, d$, and $\theta \in \mathbb{R}^p$ denotes the vector whose components are forms by collecting all entries of W_i and all components of b_i 's, and d is the total number of layers in the neural network. If $P_\rho(\cdot, \varepsilon)$ is a smoothing function of the ReLU function $(t)_+$ associated to the density ρ given by (14), then we choose

$$S(x, \theta, \varepsilon) = W_d P_\rho(\cdots P_\rho(W_2 P_\rho(W_1 x + b_1, \varepsilon) + b_2, \varepsilon) \cdots, \varepsilon) + b_d \quad (17)$$

as a smoothing function of $g(x, \theta)$. It follows from Proposition 3.2 and Lemma 3.3 that we can obtain the following.

Theorem 3.4: *Let $\tilde{f}_n(\theta, \varepsilon) = l(S(x_n, \theta, \varepsilon), y_n)$ where $l : \mathbb{R}^m \rightarrow \mathbb{R}$ is a continuously differentiable function for every $n = 1, \dots, N$. Let $\rho : \mathbb{R} \rightarrow \mathbb{R}_+$ be a piecewise continuous density function satisfying (13) and $P_\rho : \mathbb{R} \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be a smoothing function of the plus function $(t)_+$ given by (14). Then, $\tilde{f}_n(\theta, \varepsilon)$ is a smoothing function of $f_n(\theta)$ and $S_\rho(\theta, \varepsilon) := \sum_{n=1}^N l(\tilde{f}_n(\theta, \varepsilon) - y_n)$ is a smoothing function of f given in (7) with the following properties*

- (i) *For any $\theta \in \mathbb{R}^p$, $G_{S_\rho}(\theta)$ given by (10) is nonempty and bounded, and $\partial f(\theta) = G_{S_\rho}(\theta)$.*
- (ii) *If the function l is convex and nondecreasing, then $S_\rho(\cdot, \varepsilon)$ is convex for any fixed $\varepsilon > 0$.*

Remark: This theorem tells us how to construct a family of smoothing approximations to the loss functions in deep learning when ReLU is used as the activation function. The standard algorithm in deep learning is to apply the gradient descent algorithm (or more precisely, the stochastic gradient descent algorithm) with back propagation when updating the gradients (see, e.g. [13]).

4. Dynamical system inspired algorithm via smoothing approximation

In this section, using the Zang density functions introduced in [19], we consider

$$\rho(t) := \rho_Z(t) = \begin{cases} 1, & \text{if } -\frac{1}{2} \leq t \leq \frac{1}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

The smoothing Zang plus function $P_{\rho_Z}(t, \varepsilon)$ takes the form

$$P_{\rho_Z}(t, \varepsilon) = \begin{cases} t, & \text{if } t > \frac{\varepsilon}{2}, \\ \frac{1}{2\varepsilon} \left(t + \frac{\varepsilon}{2}\right)^2, & \text{if } -\frac{\varepsilon}{2} \leq t \leq \frac{\varepsilon}{2}, \\ 0, & \text{if } t < -\frac{\varepsilon}{2}. \end{cases} \quad (18)$$

We use $S(\theta, \varepsilon)$ to denote the resulting function when we replace ReLU function by $P_{\rho_Z}(t, \varepsilon)$. That is, we take $\rho = \rho_Z$ in Theorem 3.4.

Therefore, for $\varepsilon > 0$ we consider the first-order dynamical system (5) recalled below:

$$\begin{cases} \dot{\theta}(t) = -\alpha\theta(t) + \varphi(t) - \beta \nabla S(\theta(t), \varepsilon) \\ \dot{\phi}(t) = -\nabla S(\theta(t), \varepsilon) \end{cases} \quad \text{for } t \in (0, \infty) \quad (19)$$

Now we discretize (19) by using the Euler method to obtain our smoothing approximation algorithm. Given any time step $\{t_k\}$ with $t_{k+1} - t_k = \gamma_k > 0$, a small positive number. Set $\theta_k := \theta(t_k)$ and $\varphi_k := \varphi(t_k)$ for every $k \in \mathbb{N}$ for a given solution (θ, φ) (depending on $\varepsilon > 0$) of (19). One can approximate $\dot{\theta}(t_k)$ and $\dot{\phi}(t_k)$ by

$$\dot{\theta}(t_k) \simeq \frac{\theta_{k+1} - \theta_k}{\gamma_k}, \quad \dot{\phi}(t_k) \simeq \frac{\varphi_{k+1} - \varphi_k}{\gamma_k}.$$

Applying the approach of Chen [17] as described in Section 3, we can use this discretization to propose the following Smoothing Inertial Newton Algorithm (SINA) (deterministic form):

Algorithm 4.1 (Smoothing Inertial Newton Algorithm):

Step 1 Choose hyper-parameters $\alpha > 0$ and $\beta > 0$, $\sigma \in (0, 1)$, $\{\gamma_k\}_{k \in \mathbb{N}}$ ($\gamma_k > 0$), $\varepsilon_0 \in (0, 1)$, $\delta > 0$, and an initial point $(\theta_0, \psi_0) \in \mathbb{R}^p \times \mathbb{R}^p$. Set $k = 0$.

Step 2 Compute the gradient

$$g_k = \nabla_{\theta} S(\theta_k, \varepsilon_k)$$

Step 3

$$\begin{cases} \theta_{k+1} = \theta_k + \gamma_k(-\alpha\theta_k + \varphi_k - \beta g_k) \\ \varphi_{k+1} = \varphi_k - \gamma_k g_k \end{cases} \quad (20)$$

Step 4 If $\|\nabla_{\theta} S(\theta_{k+1}, \varepsilon_k)\| \geq \delta \varepsilon_k$, then set $\varepsilon_{k+1} = \varepsilon_k$; otherwise, choose $\varepsilon_{k+1} = \sigma \varepsilon_k$.

Remarks: (i) Note that Algorithm 1 (as well as Algorithm 2 stated below) is stated for general functions $S(\theta, \varepsilon)$, but when it is applied to the smoothing approximation of loss functions in machine learning (as described in Section 3 and beginning of this section), the computation of Step 2 will be where the sum rule of gradients is needed.

(ii) The computation of the sum of the gradients in every iteration is time consuming if not impossible. In practice, as in machine learning problems, one applies a stochastic procedure such as dividing the sum into many partial sums and randomly selects one block at each iteration, yielding a stochastic version of the algorithm. We postpone the treatment of the stochastic version and the study of its convergence to a future work.

To ensure the convergence of Algorithm 1, we need the following standard assumption.

Assumption 1: For every fixed $\varepsilon > 0$,

- (i) $S(\cdot, \varepsilon)$ is bounded from below;
- (ii) $S(\cdot, \varepsilon)$ is L -smooth: There exists a constant $L > 0$ such that the gradient of $S(\cdot, \varepsilon)$ satisfies

$$\|\nabla_{\theta} S(\theta_1, \varepsilon) - \nabla_{\theta} S(\theta_2, \varepsilon)\| \leq L \|\theta_1 - \theta_2\|, \quad \text{for all } \theta_1, \theta_2 \in \mathbb{R}^p. \quad (21)$$

Under the Assumption thmA, we will write $S_{\varepsilon}^* := \inf_{\theta \in \mathbb{R}^p} S(\theta, \varepsilon)$. We will also need the following assumptions on $\{\gamma_k\}_{k \in \mathbb{N}}$.

Assumption 2: The step-size sequence $\{\gamma_k\}$ satisfies the following conditions.

- (i) $0 < \gamma_k \leq \min\{\beta, 1/\alpha, \alpha/L\}$.
- (ii) $\gamma_{k+1} \leq \gamma_k$ and $\gamma_{k+1}^2 \leq \gamma_k \gamma_{k+2}$.
- (iii) $\liminf_{k \rightarrow \infty} \gamma_k > 0$.

Now, we are ready to establish the convergence of Algorithm 1 in the following theorem.

Theorem 4.1: Suppose that $S(\cdot, \varepsilon)$ satisfies Assumption thmA, and $\{\gamma_k\}$ satisfies Assumption thmB. Let $\{\theta_k\}_{k \in \mathbb{N}}$ be generated by Algorithm 1. Then every accumulation point of $\{\theta_k\}_{k \in \mathbb{N}}$ is a critical point of f .

To prove Theorem 4.1, we first establish the following result, not for a family of functions $\{S(\cdot, \varepsilon)\}_{\varepsilon > 0}$, but only for one function $S(\cdot, \varepsilon)$ with a fixed ε . For simplicity, we will drop the reference to ε and write $S(\cdot) := S(\cdot, \varepsilon)$ and $S^* := S_{\varepsilon}^*$ when ε is fixed. Consider the following simplified algorithm.

Algorithm 4.2 (When ε is fixed): Step 1 Choose hyper-parameters $\alpha > 0$ and $\beta > 0$, $\sigma \in (0, 1)$, $\{\gamma_k\}$ satisfying Assumption thmB, $\delta > 0$, and an initial point $(\theta_0, \psi_0) \in \mathbb{R}^p \times \mathbb{R}^p$. Set $k = 0$.

Step 2 Compute the gradient

$$g_k = \nabla S(\theta_k)$$

Step 3

$$\theta_{k+1} = \theta_k + \gamma_k(-\alpha\theta_k + \varphi_k - \beta g_k) \quad (22)$$

$$\varphi_{k+1} = \varphi_k - \gamma_k g_k \quad (23)$$

Step 4 $k = k + 1$ and go to Step 2 until the stop criterion is satisfied.

We remark that Algorithm 2 can be viewed as an explicit discretization of DIN given in (1) for L -smooth f . Under additional assumption of convexity or strong convexity, Attouch et al. in [9] established linear convergence. So the following result supplements

the results in [9] for the case of no convexity: it assures that the algorithm will generate a sequence on which the gradients are approaching to 0.

Theorem 4.2: Suppose that $S(\cdot) = S(\cdot, \varepsilon)$ satisfies Assumption thmA, and $\{\gamma_k\}_{k \in \mathbb{N}}$ satisfies Assumption thmB. Then, for $\{\theta_k\}_{k \in \mathbb{N}}$ generated by Algorithm 2, we have

$$\lim_{k \rightarrow \infty} \|\nabla S(\theta_k)\| = 0.$$

Proof: We first prove the theorem when $\gamma_k = \gamma$ for all k as this will make the idea of the proof more transparent than that in the non-constant case. Define the following Lyapunov function:

$$E_k = \mu(S(\theta_k) - S^*) + \|\alpha\theta_k + \varphi_k\|^2,$$

where $\mu > 0$ is a constant to be determined. We will choose μ such that E_k is nonincreasing.

First, by Assumption thmA(ii), it is known that we can obtain

$$\begin{aligned} \mu S(\theta_{k+1}) - \mu S(\theta_k) &\leq \mu \langle \nabla S(\theta_k), \Delta\theta_k \rangle + \frac{\mu L}{2} \|\Delta\theta_k\|^2 \\ &= \mu \langle g_k, \Delta\theta_k \rangle + \frac{\mu L}{2} \|\Delta\theta_k\|^2, \end{aligned} \quad (24)$$

where in the last equality, we used the notation (as given in the Algorithm 2): $g_k = \nabla S(\theta_k)$.

Next, we estimate

$$\begin{aligned} &\|\alpha\theta_{k+1} + \varphi_{k+1}\|^2 - \|\alpha\theta_k + \varphi_k\|^2 \\ &= \|\alpha(\theta_k + \Delta\theta_k) + \varphi_k + \Delta\varphi_k\|^2 - \|\alpha\theta_k + \varphi_k\|^2 \\ &= \|\alpha\Delta\theta_k + \Delta\varphi_k\|^2 + 2\langle -\alpha\theta_k + \varphi_k, -\alpha\Delta\theta_k + \Delta\varphi_k \rangle \\ &= \|\alpha\Delta\theta_k + \gamma g_k\|^2 + 2\left\langle \frac{\Delta\theta_k}{\gamma} + \beta g_k, -\alpha\Delta\theta_k - \gamma g_k \right\rangle \\ &= \alpha^2 \|\Delta\theta_k\|^2 + \gamma^2 \|g_k\|^2 + 2\alpha\gamma \langle \Delta\theta_k, g_k \rangle - \frac{2\alpha}{\gamma} \|\Delta\theta_k\|^2 \\ &\quad - 2\beta\gamma \|g_k\|^2 - 2\langle \Delta\theta_k, g_k \rangle - 2\alpha\beta \langle g_k, \Delta\theta_k \rangle \\ &= \left(\alpha^2 - \frac{2\alpha}{\gamma} \right) \|\Delta\theta_k\|^2 + \gamma(\gamma - 2\beta) \|g_k\|^2 \\ &\quad + 2(\alpha\gamma - 1 - \alpha\beta) \langle g_k, \Delta\theta_k \rangle, \end{aligned} \quad (25)$$

where for the equality in (25) we used (22) and (23). Now, adding (24) and (26), we obtain

$$\begin{aligned} E_{k+1} - E_k &\leq \mu \langle g_k, \Delta\theta_k \rangle + \frac{\mu L}{2} \|\Delta\theta_k\|^2 + \left(\alpha^2 - \frac{2\alpha}{\gamma} \right) \|\Delta\theta_k\|^2 \\ &\quad + \gamma(\gamma - 2\beta) \|g_k\|^2 + 2(\alpha\gamma - 1 - \alpha\beta) \langle g_k, \Delta\theta_k \rangle \\ &= \left(\frac{\mu L}{2} + \alpha^2 - \frac{2\alpha}{\gamma} \right) \|\Delta\theta_k\|^2 + \gamma(\gamma - 2\beta) \|g_k\|^2 \\ &\quad + (\mu + 2(\alpha\gamma - 1 - \alpha\beta)) \langle g_k, \Delta\theta_k \rangle. \end{aligned}$$

Taking $\mu = 2(1 + \alpha\beta - \alpha\gamma)$, we see that $\mu > 0$ when $\gamma > 0$ is small enough, and we obtain

$$E_{k+1} - E_k \leq \left(\frac{\mu L}{2} + \alpha^2 - \frac{2\alpha}{\gamma} \right) \|\Delta\theta_k\|^2 + \gamma(\gamma - 2\beta)\|g_k\|^2. \quad (27)$$

By making $\gamma > 0$ to be small enough, say,

$$\gamma \leq 2\beta \quad \text{and} \quad (1 + \alpha\beta - \alpha\gamma)L + \alpha^2 - \frac{2\alpha}{\gamma} \leq 0,$$

we see that the coefficients of the two terms on the right-hand side of (27) will be both non-positive. Hence $\{E_k\}$ is a nonincreasing sequence. Since S is bounded from below by Assumption (i), so is E_k . Hence, $\lim_{k \rightarrow \infty} E_k$ exists. Finally, summing in k on both sides of (27) will imply

$$\sum_{k=1}^{\infty} \|\Delta\theta_k\|^2 < \infty \quad \text{and} \quad \sum_{k=1}^{\infty} \|\nabla S(\theta_k, \varepsilon)\|^2 < \infty.$$

We get $\lim_{k \rightarrow \infty} \nabla S(\theta_k, \varepsilon) = 0$ in view of the second convergent series. This completes the proof for the case when $\gamma_k = \gamma$, a small constant.

Next, we give the proof for the general case when γ_k may vary with k (but under Assumption thmB). The proof follows the same idea but becomes messier when dealing with varying γ_k .

By Assumption thmA(i), there is a constant S^* such that $S(\theta, \varepsilon) \geq S^*$ for all θ . Define

$$E_k = \mu_k (S(\theta_k, \varepsilon) - S^*) + \gamma_k^2 \|\alpha\theta_k + \varphi_k\|^2,$$

where the sequence $\{\mu_k\}$ is to be determined.

From Assumption thmA(ii), we can obtain

$$\begin{aligned} & \mu_{k+1} (S(\theta_{k+1}, \varepsilon) - S^*) - \mu_k (S(\theta_k, \varepsilon) - S^*) \\ &= (\mu_{k+1} - \mu_k) (S(\theta_{k+1}, \varepsilon) - S^*) + \mu_k (S(\theta_{k+1}, \varepsilon) - S(\theta_k, \varepsilon)) \\ &\leq (\mu_{k+1} - \mu_k) (S(\theta_{k+1}, \varepsilon) - S^*) + \mu_k \langle \nabla_\theta S(\theta_k, \varepsilon), \Delta\theta_k \rangle + \frac{\mu_k L}{2} \|\Delta\theta_k\|^2 \\ &= (\mu_{k+1} - \mu_k) (S(\theta_{k+1}, \varepsilon) - S^*) - \frac{\mu_k}{\gamma_k} \langle \Delta\varphi_k, \Delta\theta_k \rangle + \frac{\mu_k L}{2} \|\Delta\theta_k\|^2, \end{aligned} \quad (28)$$

where in the last equality (in (27)) we employed (23). Next, we estimate

$$\begin{aligned} & \gamma_{k+1}^2 \|\alpha\theta_{k+1} + \varphi_{k+1}\|^2 - \gamma_k^2 \|\alpha\theta_k + \varphi_k\|^2 \\ &= \gamma_{k+1}^2 \|\alpha(\theta_k + \Delta\theta_k) + \varphi_k + \Delta\varphi_k\|^2 - \gamma_k^2 \|\alpha\theta_k + \varphi_k\|^2 \\ &= (\gamma_{k+1}^2 - \gamma_k^2) \|\alpha\theta_k + \varphi_k\|^2 + \gamma_{k+1}^2 \|\alpha\Delta\theta_k + \Delta\varphi_k\|^2 \\ &\quad + 2\gamma_{k+1}^2 \langle -\alpha\theta_k + \varphi_k, -\alpha\Delta\theta_k + \Delta\varphi_k \rangle \end{aligned}$$

which, by using $-\alpha\theta_k + \varphi_k = (\Delta\theta_k - \beta\Delta\varphi_k)/\gamma_k$ as implied by (22) and (23), we can continue to simplify and expand to get

$$\begin{aligned}
& \gamma_{k+1}^2 \|\alpha\theta_{k+1} + \varphi_{k+1}\|^2 - \gamma_k^2 \|\alpha\theta_k + \varphi_k\|^2 \\
&= \frac{\gamma_{k+1}^2 - \gamma_k^2}{\gamma_k^2} \|\Delta\theta_k - \beta\Delta\varphi_k\|^2 + \gamma_{k+1}^2 \|\alpha\Delta\theta_k + \Delta\varphi_k\|^2 \\
&\quad + \frac{2\gamma_{k+1}^2}{\gamma_k} \langle \Delta\theta_k - \beta\Delta\varphi_k, -\alpha\Delta\theta_k + \Delta\varphi_k \rangle \\
&= \frac{\gamma_{k+1}^2 - \gamma_k^2}{\gamma_k^2} (\|\Delta\theta_k\|^2 + \beta^2 \|\Delta\varphi_k\|^2 - 2\beta \langle \Delta\theta_k, \Delta\varphi_k \rangle) \\
&\quad + \gamma_{k+1}^2 (\alpha^2 \|\Delta\theta_k\|^2 + \|\Delta\varphi_k\|^2 - 2\alpha \langle \Delta\theta_k, \Delta\varphi_k \rangle) \\
&\quad + \frac{2\gamma_{k+1}^2}{\gamma_k} (-\alpha \|\Delta\theta_k\|^2 + \langle \Delta\theta_k, \Delta\varphi_k \rangle + \alpha\beta \langle \Delta\varphi_k, \Delta\theta_k \rangle - \beta \|\Delta\varphi_k\|^2) \\
&= \left(\frac{\gamma_{k+1}^2 - \gamma_k^2}{\gamma_k^2} + \alpha^2 \gamma_{k+1}^2 - \frac{2\alpha\gamma_{k+1}^2}{\gamma_k} \right) \|\Delta\theta_k\|^2 \\
&\quad + \left(\frac{\beta^2(\gamma_{k+1}^2 - \gamma_k^2)}{\gamma_k^2} + \gamma_{k+1}^2 - \frac{2\beta\gamma_{k+1}^2}{\gamma_k} \right) \|\Delta\varphi_k\|^2 \\
&\quad - 2 \left(\frac{\beta(\gamma_{k+1}^2 - \gamma_k^2)}{\gamma_k^2} + \alpha\gamma_{k+1}^2 - (1 + \alpha\beta) \frac{\gamma_{k+1}^2}{\gamma_k} \right) \langle \Delta\theta_k, \Delta\varphi_k \rangle. \tag{29}
\end{aligned}$$

Now, adding the inequalities obtained from (28) and (29), we obtain

$$\begin{aligned}
E_{k+1} - E_k &\leq (\mu_{k+1} - \mu_k)(S(\theta_{k+1}, \varepsilon) - S^*) - \frac{\mu_k}{\gamma_k} \langle \Delta\varphi_k, \Delta\theta_k \rangle + \frac{\mu_k L}{2} \|\Delta\theta_k\|^2 \\
&\quad + \left(\frac{\gamma_{k+1}^2 - \gamma_k^2}{\gamma_k^2} + \alpha^2 \gamma_{k+1}^2 - \frac{2\alpha\gamma_{k+1}^2}{\gamma_k} \right) \|\Delta\theta_k\|^2 \\
&\quad + \left(\frac{\beta^2(\gamma_{k+1}^2 - \gamma_k^2)}{\gamma_k^2} + \gamma_{k+1}^2 - \frac{2\beta\gamma_{k+1}^2}{\gamma_k} \right) \|\Delta\varphi_k\|^2 \\
&\quad - 2 \left(\frac{\beta(\gamma_{k+1}^2 - \gamma_k^2)}{\gamma_k^2} + \alpha\gamma_{k+1}^2 - (1 + \alpha\beta) \frac{\gamma_{k+1}^2}{\gamma_k} \right) \langle \Delta\theta_k, \Delta\varphi_k \rangle. \tag{30}
\end{aligned}$$

Taking $\mu_k = 2\gamma_k((1 + \alpha\beta)\frac{\gamma_{k+1}^2}{\gamma_k} - \frac{\beta(\gamma_{k+1}^2 - \gamma_k^2)}{\gamma_k^2} - \alpha\gamma_{k+1}^2)$, we obtain

$$\begin{aligned} E_{k+1} - E_k &\leq (\mu_{k+1} - \mu_k)(S(\theta_{k+1}, \varepsilon) - S^*) \\ &\quad + \left(\frac{\mu_k L}{2} + \frac{\gamma_{k+1}^2 - \gamma_k^2}{\gamma_k^2} + \alpha^2 \gamma_{k+1}^2 - \frac{2\alpha\gamma_{k+1}^2}{\gamma_k} \right) \|\Delta\theta_k\|^2 \\ &\quad + \left(\frac{\beta^2(\gamma_{k+1}^2 - \gamma_k^2)}{\gamma_k^2} + \gamma_{k+1}^2 - \frac{2\beta\gamma_{k+1}^2}{\gamma_k} \right) \|\Delta\varphi_k\|^2 \\ &=: A_k(S(\theta_{k+1}, \varepsilon) - S^*) + B_k\|\Delta\theta_k\|^2 + C_k\|\Delta\varphi_k\|^2. \end{aligned} \quad (31)$$

We have the following compact forms of the coefficients A_k, B_k, C_k , and μ_k .

$$\mu_k = 2\beta\gamma_k - 2\gamma_k \left(\frac{\beta\gamma_{k+1}}{\gamma_k} - \gamma_{k+1} \right) \left(\frac{\gamma_{k+1}}{\gamma_k} - \alpha\gamma_{k+1} \right), \quad (32)$$

$$B := B_k = \frac{\mu_k L}{2} + \left(\frac{\gamma_{k+1}}{\gamma_k} - \alpha\gamma_{k+1} \right)^2 - 1, \quad (33)$$

$$C := C_k = \left(\frac{\beta\gamma_{k+1}}{\gamma_k} - \gamma_{k+1} \right)^2 - \beta^2, \quad (34)$$

and finally,

$$A := A_k = \mu_{k+1} - \mu_k. \quad (35)$$

We next verify that, under Assumption thmB, we have $A \leq 0, B \leq 0, C \leq 0$, and $\mu_k \geq 0$. ■

5. Verification of $C \leq 0$

We first show that $C \leq 0$ as follows. By Assumption thmB, we know that

$$0 \leq \frac{\gamma_{k+1}}{\gamma_k} \leq 1 \quad \text{and} \quad 0 \leq \beta - \gamma_k \leq \beta. \quad (36)$$

So, we can write

$$\begin{aligned} C &= \left(\frac{\gamma_{k+1}}{\gamma_k} \right)^2 (\beta - \gamma_k)^2 - \beta^2 \\ &\leq (\beta - \gamma_k)^2 - \beta^2 \\ &= -\gamma_k(\beta - \gamma_k + \beta) \leq -\beta\gamma_k \\ &\leq 0. \end{aligned} \quad (37)$$

6. Verification of $\mu_k \geq 0$

We have

$$\mu_k \geq 0 \iff \left(\frac{\gamma_{k+1}}{\gamma_k} \right)^2 (\beta - \gamma_k)(1 - \alpha\gamma_k) \leq \beta.$$

Using (36), together with $0 \leq 1 - \alpha\gamma_k \leq 1$, we get

$$\left(\frac{\gamma_{k+1}}{\gamma_k} \right)^2 (\beta - \gamma_k)(1 - \alpha\gamma_k) \leq (\beta - \gamma_k)(1 - \alpha\gamma_k) \leq \beta.$$

7. Verification of $B \leq 0$

Now we turn to prove that $B \leq 0$. This can be seen as follows. We have

$$B = \gamma_k L \left[\beta - \left(\frac{\gamma_{k+1}}{\gamma_k} \right)^2 (\beta - \gamma_k)(1 - \alpha\gamma_k) \right] + \left(\frac{\gamma_{k+1}}{\gamma_k} \right)^2 (1 - \alpha\gamma_k)^2 - 1.$$

Hence, it suffices to verify that

$$\left(\frac{\gamma_{k+1}}{\gamma_k} \right)^2 [(1 - \alpha\gamma_k)^2 - \gamma_k L(\beta - \gamma_k)(1 - \alpha\gamma_k)] \leq 1 - \gamma_k L\beta. \quad (38)$$

Note that, from Assumption thmB(i), we have

$$\gamma_k \leq \frac{\alpha}{L}.$$

This can be written equivalently as

$$-\alpha - L\beta + L\gamma_k \leq -L\beta,$$

which implies that

$$1 - \alpha\gamma_k - \gamma_k L\beta + L\gamma_k^2 \leq 1 - \gamma_k L\beta.$$

Therefore, using Assumption thmB again, we get

$$\left(\frac{\gamma_{k+1}}{\gamma_k} \right)^2 (1 - \alpha\gamma_k) [(1 - \alpha\gamma_k) - \gamma_k L(\beta - \gamma_k)] \leq 1 - \gamma_k L\beta,$$

which is equivalent to (38).

8. Verification of $A \leq 0$

Finally, we prove that $A = \mu_{k+1} - \mu_k \leq 0$. Note that

$$\begin{aligned}
 \mu_{k+1} - \mu_k &= 2\gamma_{k+1} \left[\beta - \left(\frac{\gamma_{k+2}}{\gamma_{k+1}} \right)^2 (\beta - \gamma_{k+1})(1 - \alpha\gamma_{k+1}) \right] \\
 &\quad - 2\gamma_k \left[\beta - \left(\frac{\gamma_{k+1}}{\gamma_k} \right)^2 (\beta - \gamma_k)(1 - \alpha\gamma_k) \right] \\
 &= 2 \left[\beta(\gamma_{k+1} - \gamma_k) - \gamma_{k+1} \left(\frac{\gamma_{k+2}}{\gamma_{k+1}} \right)^2 (\beta - \gamma_{k+1})(1 - \alpha\gamma_{k+1}) \right. \\
 &\quad \left. + \gamma_k \left(\frac{\gamma_{k+1}}{\gamma_k} \right)^2 (\beta - \gamma_k)(1 - \alpha\gamma_k) \right] \\
 &= 2 [(\gamma_{k+1} - \gamma_k)(\beta - \delta_{k+1}) - \gamma_k(\delta_{k+1} - \delta_k)],
 \end{aligned}$$

where, for $k = 1, 2, 3, \dots$,

$$\delta_k := \left(\frac{\gamma_{k+1}}{\gamma_k} \right)^2 (\beta - \gamma_k)(1 - \alpha\gamma_k).$$

Since $(\beta - \gamma)(1 - \alpha\gamma)$ is a decreasing function in γ over the interval $(0, \min\{\frac{1}{\beta}, \frac{1}{\gamma}\}]$, together with the fact that (by Assumption thmB)

$$0 \leq \gamma_{k+1} \leq \gamma_k \quad \text{and} \quad \frac{\gamma_{k+2}}{\gamma_{k+1}} \geq \frac{\gamma_{k+1}}{\gamma_k} \geq 0,$$

we can write $\gamma_{k+1} \leq \gamma_k$ and $\delta_k \leq \delta_{k+1} \leq \beta$. Then it follows that

$$\begin{aligned}
 \mu_{k+1} - \mu_k &= 2 [(\gamma_{k+1} - \gamma_k)(\beta - \delta_{k+1}) - \gamma_k(\delta_{k+1} - \delta_k)] \\
 &\leq 2(\gamma_{k+1} - \gamma_k)(\beta - \delta_{k+1}) \leq 0.
 \end{aligned}$$

Hence, $\{E_k\}$ is non-increasing. Since it is also bounded from below, it follows that the limit $\lim_{k \rightarrow \infty} E_k$ exists. Let E^* denote the limit value: $E^* = \lim_{k \rightarrow \infty} E_k$.

Summing the two sides for $k = 1$ to ∞ in (31), we obtain

$$\sum_{k=1}^{\infty} -A_k(S(\theta_{k+1}, \varepsilon) - S^*) + \sum_{k=1}^{\infty} -B_k \|\Delta \theta_k\|^2 + \sum_{k=1}^{\infty} -C_k \|\Delta \varphi_k\|^2 \leq E^* - E_1. \quad (39)$$

In particular, using (37), we know that $-C_k \geq \beta\gamma_k$. Thus the last series above will yield

$$\sum_{k=1}^{\infty} \gamma_k \|\Delta \varphi_k\|^2 \leq E^* - E_1. \quad (40)$$

Noting that $\Delta \varphi_k = -\gamma_k \nabla S(\theta_k, \varepsilon)$ and $\lim_{k \rightarrow \infty} \gamma_k > 0$ (as implied by Assumption thmB), we see that inequality (40) above would imply that

$$\lim_{k \rightarrow \infty} \|\nabla S(\theta_k, \varepsilon)\| = 0.$$

This completes the proof.

Remarks: (1) When $\gamma_k \equiv \gamma$, a constant, then Assumption thmB holds.

- (2) It is easy to see that Assumption thmB(i) and (ii) imply that $\lim_{k \rightarrow \infty} \gamma_k$ exists. With Assumption thmB (iii), we know that this limit must be positive and $\lim_{k \rightarrow \infty} \frac{\gamma_{k+1}}{\gamma_k} = 1$.
- (3) The convergence implied by inequality (40) is stronger than what is just in the proof. Indeed, we have

$$\sum_{k=1}^{\infty} \|\nabla S(\theta_k, \varepsilon)\|^2 < \infty,$$

which implies the rate of convergence as $\|\nabla S(\theta_k, \varepsilon)\|$ approaches to 0 could not be arbitrarily slow.

- (4) The inequality given in (39) also implies the convergence of $\sum_{k=1}^{\infty} \|\Delta \theta_k\|^2$ as one can verify that, under Assumption thmB, we have $\lim_{k \rightarrow \infty} B_k > 0$.

We now turn to the proof of Theorem 4.1, which constitutes the main result of this paper.

Proof: Set $\mathcal{E} := \{k \in \mathbb{N} \mid \varepsilon_{k+1} = \sigma \varepsilon_k\}$. Then, by Theorem 4.2, $\mathcal{E}$ must contain infinitely many elements. Thus, $\varepsilon_k \rightarrow 0$ as $k \rightarrow \infty$, and $\lim_{k \rightarrow \infty} \nabla S(\theta_k, \varepsilon_k) = 0$. This implies that, if θ is a limit point of the sequence $\{\theta_k\}$, then $0 \in G_S(\theta)$ by definition. Since $S(\theta, \varepsilon)$ is chosen to be a smoothing approximation of $f(\theta)$, we have $0 \in \partial f(\theta)$ by the gradient consistency. ■

Disclosure statement

No potential conflict of interest was reported by the author(s).

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